

RESEARCH ARTICLE

Surface-Engineered LSCF Electrode via $\text{PrNi}_{0.7}\text{Co}_{0.3}\text{O}_{3-\delta}$ Infiltration for Superior Performance and Stability in Reversible Protonic Ceramic Cells

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ABSTRACT

The reversible protonic ceramic cells (R-PCCs) have been extensively researched for power generation and hydrogen production at low and intermediate temperatures. However, sluggish kinetics of the air electrode at low temperatures, coupled with the lack of durability under high steam environments, hinder commercialization. In this work, we developed an air electrode with high catalytic activity and durability by surface-modifying conventional $\text{La}_{0.6}\text{Sr}_{0.4}\text{Co}_{0.2}\text{Fe}_{0.8}\text{O}_{3-\delta}$ (LSCF) using single-layer perovskite $\text{PrNi}_{0.7}\text{Co}_{0.3}\text{O}_{3-\delta}$ (PNC-73) through solution infiltration. PNC-73 is a triple-conducting oxide (TCO) that extends proton conduction paths from the triple-phase boundary (TPB) to the bulk surface, unlike conventional air electrodes. Additionally, PrO_2 phase formation during infiltration creates oxygen vacancies, improving catalytic activity. PNC-73-coated LSCF exhibited a polarization resistance (R_p) of $0.30 \, \Omega \cdot \text{cm}^2$ at 650°C under both dry/humidified conditions, four times lower than the bare LSCF. Furthermore, the infiltration reduced the degradation rate to $6.5 \times 10^{-4} \, \Omega \cdot \text{cm}^2 \cdot \text{h}^{-1}$, nearly half that of bare LSCF ($12.1 \times 10^{-4} \, \Omega \cdot \text{cm}^2 \cdot \text{h}^{-1}$) under a 30 v.% H_2O environment at 650°C . The single cell (Ni-BZCYYb/BZCYYb/LSCF-PNC-73) demonstrated a power density of $500 \, \text{mW} \cdot \text{cm}^{-2}$ at 700°C in fuel cell mode and a current density of $2.80 \, \text{A} \cdot \text{cm}^{-2}$ at 1.3 V in electrolysis mode, higher than bare LSCF air electrodes. This research provides direction to mitigate major conventional air electrodes issues in R-PCCs, highlighting commercial viability.

1 | Introduction

The increasing energy demand due to the rapid advancement in technology, transportation, and industry is putting more effort into incorporating clean energy sources. This shift is important to address the environmental concerns caused by conventional energy sources, particularly fossil fuels [1, 2]. Over the last few decades, the share of renewable and sustainable energy has been steadily increasing, which presents challenges in efficiently storing excessive energy during peak shaving hours [3]. Solid oxide

fuel cells (SOFCs) are a promising energy conversion technology with the capability of hydrogen production and energy generation in reversible operating modes. SOFCs exhibit high power efficiency, fuel flexibility, good stability, low emissions, and relatively low cost compared to other energy conversion devices [4, 5]. Based on ion conduction, SOFCs are categorized into oxygen-ion conducting SOFCs (O-SOFCs) and proton-conducting SOFCs (H-SOFCs, also known as protonic ceramic fuel cells (PCFCs) [6]. The PCFCs have gained more attention over the last decade due to their versatile properties compared to O-SOFCs [7–9]:

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(1) The activation energy for proton diffusion is lower than that for oxygen-ion diffusion, which makes PCFCs suitable for operation at lower temperatures (700°C–450°C), resulting in enhanced mechanical stability, shorter startup times, easier sealing, and reduced costs; [10] (2) The byproduct water produced at the cathode prevents dilution of fuels and is beneficial for nickel-based anodes, helping to avoid oxidation and leading to a longer lifespan; [11, 12] (3) The low electronic conductivity of protonic electrolytes boosts the power output and efficiency of the fuel cell at low temperatures [13], and (4) The low and intermediate operating temperature increase protonation in the fuel cell, which enhances proton conductivity and overall performance of PCFCs [14]. The reduction in the operating temperature of the fuel cell offers benefits, as discussed above, but it introduces sluggish kinetic reactions, specifically the oxidation-reduction reaction (ORR) and oxygen evolution reaction (OER) during fuel cell and electrolysis modes [15, 16].

The air electrodes play a crucial role in the performance of fuel cells in both operating modes at intermediate and low temperatures. At present, the large-scale deployment of reversible protonic ceramic cells (R-PCCs) requires the development of air electrodes with high catalytic activity and robust durability at intermediate and low temperatures, under dry air and high-steam environments [17–19]. Many reported air electrode materials, including perovskite-structured mixed ionic and electronic conductors (MIECs) like $\text{La}_{0.6}\text{Sr}_{0.4}\text{Co}_{0.2}\text{Fe}_{0.8}\text{O}_{3-\delta}$ (LSCF), $\text{Ba}_{0.5}\text{Sr}_{0.5}\text{Co}_{0.8}\text{Fe}_{0.2}\text{O}_{3-\delta}$ (BSCF), and $\text{Sr}_2\text{Fe}_{1.5}\text{Mo}_{0.5}\text{O}_{6-\delta}$ (SFM), offer enhanced electrochemical performance but are optimized for O-SOFCs [20, 21]. In the case of R-PCCs, there is a need to introduce proton conduction in the air electrode to extend the reaction sites from the TPB to the bulk air electrode surface, thus improving the performance of the fuel cell [22]. There are two possible ways to create highly efficient air electrodes for R-PCCs capable of conducting electrons, oxygen ions, and protons: one is to develop new air electrode materials, such as TCOs, and the second is to modify conventional MIEC electrodes by incorporating TCO materials through surface modification techniques. The development of new air electrode materials with electron, oxygen-ion, and proton conduction capabilities is complex due to the challenge of creating sufficient defects for proton conduction and ensuring enough oxygen vacancies that can easily hydrate [23, 24]. Numerous researchers have reported the development of TCO air electrodes with perovskite structures that exhibited high catalytic activity and robust stability which resulted in high power and current densities, such as $\text{PrBa}_{0.5}\text{Sr}_{0.5}\text{Co}_2\text{O}_{5+\delta}$ (PBSC) [25], $\text{NdBa}_{0.5}\text{Sr}_{0.5}\text{Co}_{1.5}\text{Fe}_{0.5}\text{O}_{5+\delta}$ (NBSC) [24], $\text{PrBa}_{0.8}\text{Ca}_{0.2}\text{Co}_2\text{O}_{5+\delta}$ (PBCC) [26], $\text{PrBaCo}_{1.6}\text{Fe}_{0.2}\text{Nb}_{0.2}\text{O}_{5+\delta}$ (PBCFN) [27], $\text{PrBa}_{0.8}\text{Ca}_{0.2}\text{Fe}_{1.8}\text{Ce}_{0.2}\text{O}_{6-\delta}$ (PBCFC) [28], $\text{PrNi}_{0.5}\text{Co}_{0.5}\text{O}_{3-\delta}$ (PNC) [3], and $\text{PrNi}_{0.7}\text{Co}_{0.3}\text{O}_{3-\delta}$ (PNC) [29]. However, the synthesis complexities, the high cost of Cobalt (Co), and the unsustainability and expense of Praseodymium (Pr) hinder the commercialization of these new air electrodes. Another approach to extending the reaction sites from the TPB to the bulk electrode surface is the surface modification impregnation/infiltration of conventional air electrodes using TCO materials, which is a scalable and cost-effective method. In recent years, many researchers have reported on the infiltration of conventional MIEC air electrodes (specifically LSCF: the most widely used air electrode for O-SOFC). In addition to extending the reaction

sites, the infiltration of TCO materials also mitigates other core issues associated with LSCF air electrodes, such as high polarization resistance due to sluggish kinetic reactions and low stability caused by surface Strontium (Sr) segregation at low and intermediate temperatures [30, 31]. For example, Niu et al. [32] reported the development of a $\text{Pr}_{1-x}\text{Ba}_x\text{CoO}_{3-\delta}$ multiphase catalyst for LSCF modification through infiltration, which showed a reduction in polarization resistance by a factor of 25 and improved stability under high steam environments compared to bare LSCF. This resulted in higher power and current densities in both fuel cell and electrolysis modes. Lei et al. [33] reported the $\text{PrNi}_{0.5}\text{Mn}_{0.5}\text{O}_{3-\delta}$ and PrO_x infiltrated LSCF air electrode for PCFCs, which showed a 2.5-fold increase in peak power density compared to bare LSCF at 600°C. Other composite air electrodes have also been recently reported, such as Pr_6O_{11} -infiltrated LSM-BZY [34], PrO_x -infiltrated $\text{Ba}_{0.95}\text{La}_{0.05}\text{Fe}_{0.8}\text{Zn}_{0.2}\text{O}_{3-\delta}$ - $\text{BaZr}_{0.1}\text{Ce}_{0.7}\text{Y}_{0.1}\text{Yb}_{0.1}\text{O}_{3-\delta}$ (BLFZ-BZCYyb) [35], and $\text{Pr}_{0.2}\text{Ce}_{0.8}\text{O}_{2-\delta}$ -infiltrated SFM air electrodes [20], beside the infiltrated LSCF.

In this work, the single perovskite material $\text{PrNi}_{0.7}\text{Co}_{0.3}\text{O}_{3-\delta}$ (PNC-73) is used to modify the LSCF air electrode through a solution infiltration method. The PNC-73-coated LSCF air electrode demonstrates enhanced catalytic activity and robust stability at intermediate temperatures under dry air and high steam environments. The PNC-73 infiltration results in the formation of two phases: one is PrO_2 , which creates oxygen vacancies, and the other is PNC-73, which extends proton conduction from the TPB to the bulk electrode surface, leading to improved overall performance. The PNC-73-coated LSCF showed a polarization resistance of 0.3 Ω at 650°C under dry and 3 v.% H_2O humidified environments, which is four times lower than that of the bare LSCF air electrode. The infiltration of PNC-73 on LSCF decreases the degradation rate to $6.5 \times 10^{-4} \Omega \cdot \text{cm}^2 \cdot \text{h}^{-1}$, which is almost half of that of bare LSCF ($12.1 \times 10^{-4} \Omega \cdot \text{cm}^2 \cdot \text{h}^{-1}$) under a 30 v.% H_2O humidified environment at 650°C. The single cell based on PNC-73-coated LSCF (Ni-BZCYyb/BZCYyb/LSCF-PNC-73) exhibits remarkable performance with a power density of 500 $\text{mW} \cdot \text{cm}^{-2}$ at 700°C in fuel cell mode and a current density of 2.8 $\text{A} \cdot \text{cm}^{-2}$ at 1.3 V in electrolysis mode, which is higher than that of cells with bare LSCF air electrodes. The density functional theory (DFT) study further explores the kinetic reactions (ORR and OER) on pristine and defective PNC structures, and also analyzes and compares the oxygen vacancy formation energy of LSCF, PNC, and PrO_2 materials.

2 | Results and Discussion

2.1 | Material Design, Structural Characterization, and Proposed Reaction Mechanism

The reaction mechanism of bare LSCF and PNC-73 infiltrated LSCF air electrodes in R-PCCs is illustrated in Figure 1a. The proton (H^+) reaction sites in cells with bare LSCF air electrodes are only limited to the electrolyte/air-electrode interface because LSCF is MIEC material with high ionic and electronic conduction, but poor proton conduction. However, the infiltration of the PNC-73 solution, which is a triple conducting oxide material, extends the reaction sites for H^+ from the electrolyte/air-electrode interface to the whole air-electrode surface and creates oxygen

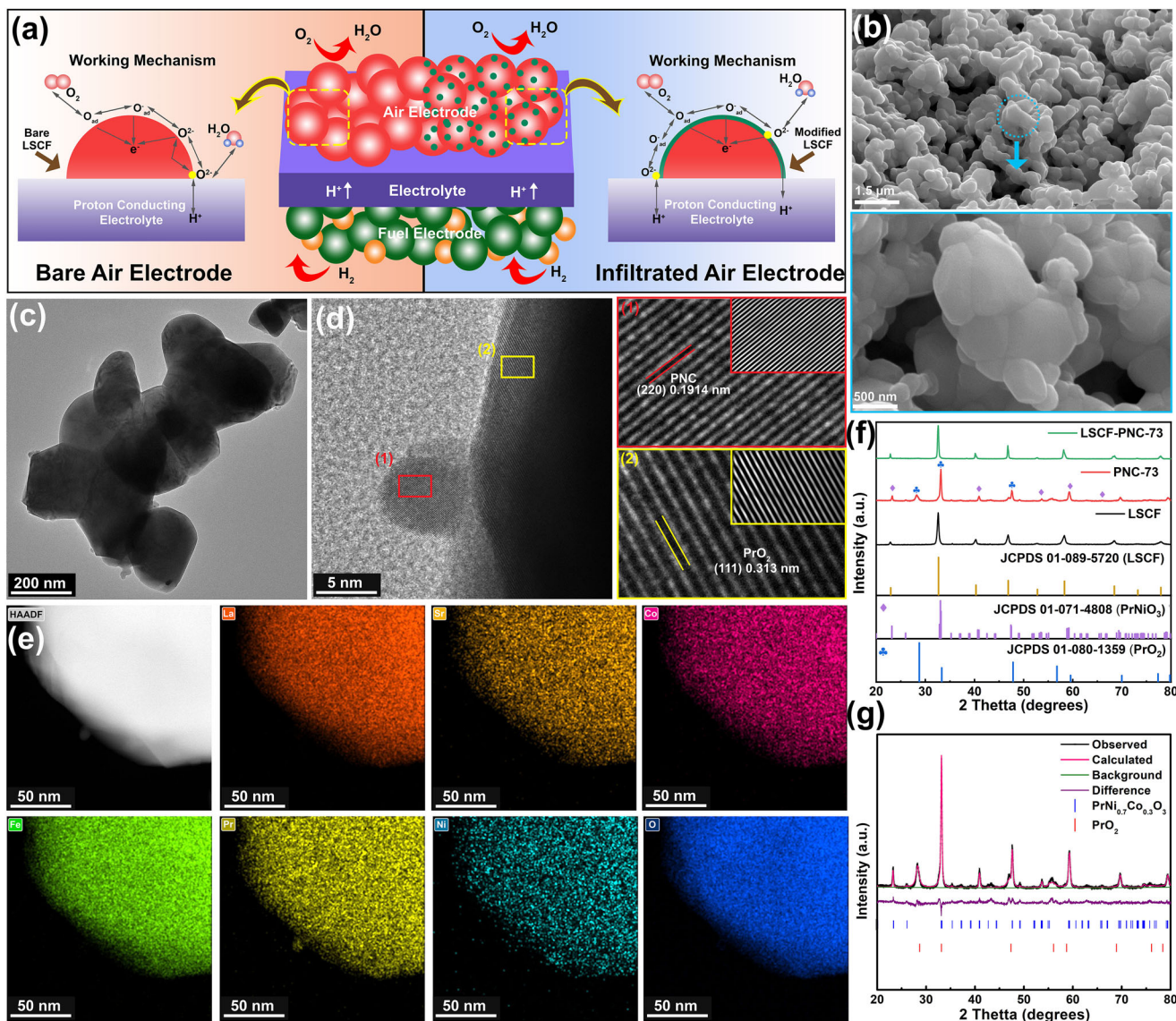


FIGURE 1 | (a) Schematic illustration of the reaction mechanism of bare LSCF and PNC-73 coated LSCF air electrodes, (b,c) SEM images and TEM image of PNC-73 coated LSCF air electrode, respectively, (d) HR-TEM image and phase structures of PNC-73 coated LSCF air electrode, (e) HAADF image with corresponding elemental mapping of PNC-73 coated LSCF air electrode, (f) XRD profiles of LSCF, PNC-73 coated LSCF, and PNC-73 powder obtained after sintering the as-developed infiltrate solution of PNC-73 precursor at 700°C, and (g) Rietveld refinement profile of PNC-73 powder.

vacancies. When PNC-73 material was infiltrated on the LSCF air electrode, two phases related to PrO₂ and PNC-73 appeared. The PrO₂ is responsible for creating oxygen vacancies, which result in accelerating the oxygen adsorption and surface exchange, whereas the PNC-73 is responsible for extending the path of proton conduction from the electrolyte/air-electrode interface to the entire air-electrode surface. The enhanced oxygen vacancies and proton pathways with the infiltration of PNC-73 material result in an improvement in the electrochemical performance of R-PCFC. The surface morphology and structure of PNC-73 coated are revealed from the scanning electron microscopy (SEM) and transmission electron microscopy (TEM) images. Figure 1b shows the SEM image of the PNC-73 coated LSCF air electrode, which depicts the conformational coating of the infiltrated solution of PNC-73 on the LSCF base electrode. The nanoparticles of PNC-73 are uniformly coated on the LSCF air electrode after co-firing at 700°C for 2 h. Figure S1 shows the SEM images and

corresponding EDS profiles of PNC-73 coated LSCF air electrodes with different infiltrate solution loadings (0, 5, 10, and 15 μ L). Figure 1c shows the TEM image of PNC-73 infiltrated LSCF air electrode, which also depicts the conformational coating of PNC-73 on the LSCF air electrode. To further explore the detailed insight into the structure and formation of phases in PNC-73 coated LSCF, high-resolution transmission electron microscopy (HR-TEM) is performed, as given in Figure 1d. There are two-phase structures observed in HR-TEM analysis of PNC-73 coated LSCF air electrodes, which are related to PrO₂ and PNC-73 materials. Figure 1(d-1) shows the lattice d-spacing of 0.313 nm, corresponding to the (111) facets of the Fm-3m group of the cubic structure of PrO₂. Moreover, Figure 1(d-2) shows the lattice with a d-spacing of 0.1914 nm, corresponding to the (220) facets of the *Pnma* group of the orthorhombic structure of PNC. The high-angle annular dark field (HAADF) image of PNC-73 coated LSCF with corresponding elemental mapping depicts the

conformal coating of the infiltrated solution and the presence of PrO_2 and PNC-73 materials, as shown in Figure 1e. The X-ray diffraction (XRD) analysis of the LSCF air electrode with and without infiltration of PNC-73 material is also analyzed to further evidence the formation of PrO_2 and PNC-73 phases. Figure 1f shows the XRD profiles of LSCF, PNC-73 infiltrated LSCF, and PNC-73 powder. The XRD profiles of LSCF and PNC-73 infiltrated LSCF look similar; no peaks related to PNC-73 appeared in the XRD pattern of PNC-73 coated LSCF air electrode material, which is owing to the low loading of the infiltrate solution. The PNC-73 material for XRD analysis is obtained after drying the infiltrate solution in an oven at 80°C for 24 h, followed by sintering at 700°C for 2 h in a furnace. The XRD pattern of the as-prepared PNC-73 powder shows diffraction peaks from PNC-73 and PrO_2 phases. Figure 1g shows the Rietveld refinement of the XRD data of PNC-73 powder. For PNC-73, an orthorhombic perovskite structure with space group $Pnma$ is calculated with lattice parameters $a = 5.384 \text{ \AA}$, $b = 5.352 \text{ \AA}$, and $c = 7.588 \text{ \AA}$. Whereas for PrO_2 , the cubic structure with space $Fm-3m$ is calculated with lattice parameter $a = b = c = 5.373 \text{ \AA}$. From morphological and X-ray diffraction analysis, it is observed that the PNC-73 material is conformably coated on the LSCF skeleton and forms a PNC-73 perovskite phase and PrO_2 secondary phase, which further results in improving the performance of R-PCFC with the as-reported infiltrated air electrode owing to the formation of oxygen vacancies and extended pathways for H^+ conduction.

X-ray photoelectron spectroscopy (XPS) was used to investigate the surface composition and chemical states of the LSCF and PNC-73 coated LSCF air electrodes. The XPS spectra shown in Figure S2a reveal the presence of all constituent elements from the LSCF and PNC materials. The XPS analysis of oxygen on LSCF before and after infiltration is presented in Figure S2b. The O 1s spectra show four oxygen species: adsorbed hydroxide (OH) at 533.3 eV , adsorbed oxygen (O_{ad}) at 531.0 eV , $\text{O}_2^{2-} / \text{O}^-$ at 529.4 eV , and lattice oxygen (O_{lat}) at 528.8 eV [36, 37]. The oxygen vacancy concentration is estimated by the ratio $\text{O}_{\text{ad}} / \text{O}_{\text{lat}}$ [38, 39]. The value of $\text{O}_{\text{ad}} / \text{O}_{\text{lat}}$ ratio before and after infiltration of PNC-73 on LSCF is 0.78 and 0.89, respectively. Figure S2c shows the Co 2p XPS spectra of LSCF and PNC-73 coated LSCF air electrode. The mixed valence states of Co^{2+} (781.2 eV for Co $2p_{3/2}$ and 796.5 eV for Co $2p_{1/2}$) and Co^{3+} (780.2 eV for Co $2p_{3/2}$ and 795.1 eV for Co $2p_{1/2}$) are observed during analysis [40, 41]. The $\text{Co}^{3+}/\text{Co}^{2+}$ ratio significantly decreases from 1.06 to 0.98 after PNC-73 infiltration onto LSCF, which depicts the decrease in valence state of Co after infiltration. This reduction in valence state of Co signifies the loss of some electrons. These lost electrons are transferred to the lattice oxygen, resulting in a decrease in the number of oxygen atoms and the formation of oxygen vacancies [42]. Figure S2d shows the Fe 2p XPS spectra of LSCF and PNC-73 coated LSCF air electrodes, revealing the presence of mixed valence states of Fe. The peaks corresponding to Fe^{3+} are observed at (723.2 and 710.1 eV), whereas the peaks for Fe^{4+} appear at (725.1 and 711.7 eV) [43, 44]. The Fe^{3+} content increases from 40% to 43.4% after PNC-73 infiltration. This increment indicates a reduction in the valence state of Fe, which is compensated by the formation of oxygen vacancies. The Sr 3d XPS spectra of LSCF and PNC-73 coated LSCF air electrode are shown in Figure S2e. The $\text{Sr}_{\text{surface}}$ peaks appear at (132.47 and 134.55 eV), while the $\text{Sr}_{\text{lattice}}$ peaks appear at (131.23 and 133.8 eV) [45]. The results show that the $\text{Sr}_{\text{surface}}$ of LSCF decreased from 53% to 50% after PNC-

73 infiltration on LSCF, which is attributed to the ability of the PrO_2 phase to suppress Sr surface segregation [46]. Figure S2f shows the La 3d XPS spectra, which show no significant shifts in the peaks before and after infiltration of PNC-73. The XPS spectra of Pr and Ni from PNC-73 coated LSCF air electrode are shown in Figure S2g,h, respectively. The creation of oxygen vacancies in the LSCF air electrode upon infiltration with the PNC-73 perovskite material was further investigated by electron paramagnetic resonance (EPR). The high intensity peaks in the EPR profile at $g = 2$ demonstrate a higher concentration of oxygen vacancies [47, 48]. Figure S2i shows the EPR spectra of the bare LSCF and PNC-73 coated LSCF air electrodes. The PNC-73 coated LSCF air electrode exhibits a high peak intensity compared to the bare LSCF at $g = 2$, indicating an increase in oxygen vacancies in LSCF following PNC-73 infiltration.

2.2 | Electrochemical Optimization and Performance Enhancement

The infiltration of the catalyst on the air electrode enhances the catalytic activity of the air electrode, which further results in an increase in the performance of R-PCFC. The co-firing of the catalyst-infiltrated air electrode at a certain temperature (typically from 600°C to 900°C) for a specific period (2 to 5 h) has a great influence on the catalytic activity of the infiltrated air electrode. The co-firing of infiltrated air electrodes serves to activate the catalyst, bonding with the air electrode skeleton, optimizing the micro-structure of the air electrode, ensuring the removal of impurities or residuals, and enhancing durability. To consider the influence of this important factor on the overall performance of the infiltrated air electrode, initially, the catalyst with composition $\text{PrNi}_{0.5}\text{Co}_{0.5}\text{O}_{3-\delta}$ (PNC-55) was infiltrated on LSCF air electrode and co-fired at different temperatures (800°C , 700°C , and 600°C) for 2 h to optimize the co-fire temperature for the rest of the electrochemical analysis. Figures S3 and S4 show the EIS profiles and temperature dependence of R_p graphs of symmetrical cells with PNC-55 coated LSCF air electrodes co-fired at 800°C , 700°C , and 600°C for 2 h under dry air and 3 v.% H_2O humidified air environment at different operation temperatures (600°C to 450°C). It is observed that the symmetrical cells co-fired at 700°C depict the R_p values of 0.65, 1.39, 2.76, and $6.70 \Omega\text{cm}^2$ under a dry air environment and 0.51, 0.97, 2.01, and $5.10 \Omega\text{cm}^2$ under 3 v.% H_2O humidified air environment at 600, 550, 500 and 450°C operating temperature, respectively. These R_p values are less as compared to the symmetrical cells co-fired at 800°C and 600°C temperatures. This analysis reveals that at 700°C co-fire temperature, the catalyst made proper bonding with the LSCF skeleton and activated robustly. After optimizing the co-fire temperature, different compositions of $\text{PrNi}_x\text{Co}_{1-x}\text{O}_{3-\delta}$ (where $x = 0.1, 0.3, 0.5, 0.7$, and 0.9) catalysts were prepared and infiltrated on the LSCF air electrode skeleton for optimizing the catalyst composition that gives the minimum value of R_p under dry air and 3 v.% H_2O humidified air environment at different operation temperatures (650°C to 450°C). Figures S5 and S6 and Figure 2a shows the EIS profiles and temperature dependence of R_p graphs of symmetrical cells based on LSCF air electrodes infiltrated with different compositions of $\text{PrNi}_x\text{Co}_{1-x}\text{O}_{3-\delta}$ (PNC, where $x = 0.1, 0.3, 0.5, 0.7$ and 0.9) catalysts under dry air and 3 v.% H_2O humidified air environment at different operation temperatures (650°C to 450°C), respectively. The LSCF symmetrical

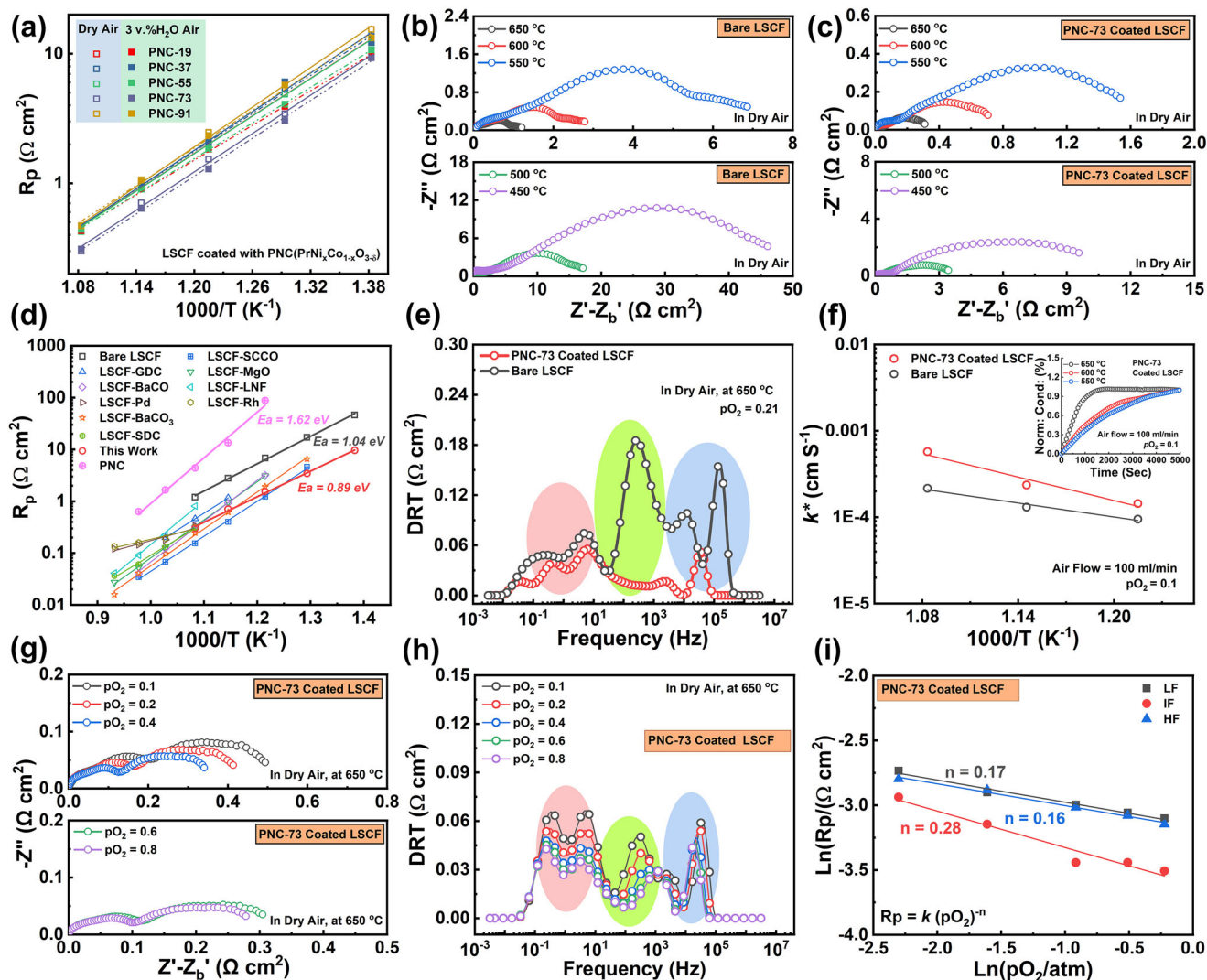


FIGURE 2 | (a) Temperature dependence of R_p of the different composition of $\text{PrNi}_x\text{Co}_{1-x}\text{O}_{3.5}$ coated LSCF air electrode, (b,c) Nyquist plots showing the EIS analysis of bare LSCF and PNC-73 coated LSCF air electrodes at different temperatures under dry air environment, respectively, (d) Temperature dependence R_p of the bare LSCF, PNC-73 coated LSCF and pure phase PNC-55 air electrodes at different temperature. The R_p values of LSCF coated with different materials reported in the literature are also included for comparison, [50, 57–64] (e) DRT profiles of the EIS analysis of bare LSCF and PNC-73 coated LSCF air electrodes observed under a dry air environment at 650 °C, (f) Temperature dependence of k^* of the bare LSCF and PNC-73 coated LSCF bars at 650–550 °C under a dry air environment (the insert image is the normalized conductivity graph of the PNC-73 coated LSCF), (g) Nyquist plots showing the EIS analysis of PNC-73 coated LSCF air electrode observed under a different partial pressure of oxygen at 650 °C, (h) DRT profiles of the EIS analysis of PNC-73 coated LSCF air electrode observed under a different partial pressure of oxygen at 650 °C, and (i) R_p respond to the change in partial pressure of oxygen for PNC-73 coated LSCF air electrode at 650 °C.

cells infiltrated with PNC-73 catalyst solution shows R_p values of 0.30, 0.70, 1.54, 3.40 and 9.50 $\Omega\text{-cm}^2$ under dry air environment and 0.30, 0.64, 1.29, 3.04, and 9.29 $\Omega\text{-cm}^2$ under 3 v.% H_2O humidified air environment at 650, 600, 550, 500, and 450 °C operating temperature, respectively. The R_p values of the LSCF symmetrical cells infiltrated with PNC-73 are significantly smaller as compared to the rest of the composition of PNC. Along with optimizing the co-fire temperature and composition of PNC infiltrate, the loading amount of infiltrate solution on the LSCF air electrode is also important. For this purpose, three different loading amounts of PNC-73 were infiltrated on the LSCF air electrodes and analyzed under dry air and 3 v.% H_2O humidified air environments at different operating temperatures, as given in Figures S7 and S8. The symmetrical cells based on LSCF air

electrodes infiltrated with 5, 10, and 15 μL PNC-73 catalyst solution show no big difference in R_p values, such as at 650 °C the R_p values of symmetrical cells with 5 and 15 μL infiltrate solution depicted the same values (0.42 $\Omega\text{-cm}^2$), whereas with 10 μL the R_p is 0.40 $\Omega\text{-cm}^2$. From this analysis, it is concluded that 5 μL is optimal for the rest of the experimental analysis without affecting the R_p values. From the aforementioned analysis and discussion, the co-fire temperature, composition, and loading amount of catalyst solution are optimized to 700 °C, PNC-73, and 5 μL values, respectively, which are finalized for the rest of the experimental analysis.

Figure 2b,c and Figures S9 and S10 show the EIS profiles of symmetrical cells with PNC-73 coated LSCF and bare LSCF

air electrodes under dry air and 3 v.% H_2O humidified air environment at 650, 600, 550, 500, and 450°C, respectively. The R_p values for instance at 650°C with bare LSCF are 4-folded ($1.2 \Omega \text{ cm}^2$) as compared to PNC-73 coated LSCF air electrodes, which show $0.30 \Omega \text{ cm}^2$. The four times smaller R_p values with coated LSCF result in higher catalytic performance. For comparison, EIS analysis of symmetrical cells based on pure phase PNC air electrode was also observed under dry air environment at different temperatures, which resulted in higher R_p values compared to bare LSCF and PNC-73 coated LSCF air electrodes. The EIS profiles of symmetrical cells with pure phase PNC air electrode is given in Figure S11. The comparison of temperature dependence R_p of the PNC-73 coated LSCF air electrode with bare LSCF, pure phase PNC, and previously reported LSCF air electrodes coated with different catalysts is given in Figure 2d. Table S1 shows the details about the configuration of symmetrical cells based on LSCF air electrode coated with different materials reported in the literature. Most of the previously reported work analyzed the modified air electrodes at high or intermediate temperatures, given in the range from (850°C to 550°C), however, in our work, we analyzed up to 450°C, and the R_p values of our reported work are lower than those of most previously reported catalysts. The activation energy for the bare LSCF and pure phase PNC air electrode are 1.04 and 1.62 eV, respectively, which are higher as compared to the PNC-73 coated LSCF 0.89 eV, which reflects the capability of the PNC-73 coated LSCF air electrode to operate at low temperatures. These activation energy values (1.04 and 0.89 eV) correspond exclusively to the optimized bare LSCF and PNC-73 coated LSCF air electrodes as given in Figure 2d. Meanwhile, the 0.74 – 0.93 eV range of activation energies reported in Figures S3 and S4 belongs to the preliminary PNC-55 coated LSCF air electrode samples, where the variation arises from different co-firing temperatures (800, 700, and 600°C) used during initial heat-treatment optimization. The interpretation of EIS graphs of symmetrical cells with respect to frequency arcs helps in a better understanding of the kinetic reactions involved in obtaining the distribution of relaxation time (DRT) data from the EIS profiles. Figure 2e shows the DRT profiles of symmetrical cells with bare LSCF and PNC-73 coated LSCF air electrodes, observed at 650°C under a dry air environment. The DRT graphs are divided into three regions: low frequency region (LF, <10 Hz), intermediate frequency region (IF, 10 Hz to 2 kHz), and high frequency region (HF, >2 kHz). These frequency regions LF, IF, and HF are associated with the gas diffusion process, oxygen surface exchange process, and charge transfer processes [32, 42, 49]. The smaller peaks in IF and HF regions of PNC-73 coated LSCF air electrode graph as compared to the bare LSCF indicate enhanced surface oxygen exchange and charge transfer process. These results of DRT are consistent with the electrical conductivity relaxation (ECR) measurements of LSCF and PNC-73 coated LSCF air electrode materials as shown in Figures 2f and S12. The ECR measurements further give information about the surface exchange co-efficient (k) values, which reflect the catalytic activity performance ability of the materials. The k value of PNC-73 coated LSCF is $5.73 \times 10^{-4} \text{ cm s}^{-1}$, which is 2.5 times as compared to the k value of bare LSCF ($2.16 \times 10^{-4} \text{ cm s}^{-1}$) at 650°C, indicating the enhanced catalytic activity after infiltration of PNC-73 into the LSCF air electrode.

To further understand the surface reaction of the as-reported air electrode, the EIS profiles and corresponding DRT graphs

of R-PCFC cells based on bare LSCF and PNC-73 coated LSCF air electrodes were obtained under different partial pressures of oxygen ($p\text{O}_2$) at 650°C as shown in Figures S13 and S14 and Figure 2g,h, respectively. The DRT graphs were divided into three frequency regions (LF, IF, and HF), which are associated to the gas diffusion process, oxygen surface exchange process, and charge transfer processes. The peaks in IF and HF regions are sensitive to the $p\text{O}_2$ variation for both bare and coated air electrodes. Moreover, from DRT graphs, a significant reduction of peaks in IF and HF regions is observed as compared to the bare LSCF air electrode, which reflects the influence on oxygen surface exchange and charge transfer processes. The relation between the R_p and $p\text{O}_2$ is further evaluated by the function $R_p = k(p\text{O}_2)^{-n}$; [50–52] where n is the reaction order (the determined rate step) with respect to $p\text{O}_2$ and k is the constant as given in Figure 2i. The value of n for the HF range is 0.16, which is associated with the reduction of O^- to O^{2-} at TPB with the transfer of electrons [49, 53]. The n value for the IF range is 0.28, which is related to the diffusion of O^- from the electrode surface to the TPB [54, 55]. For LF range the n value is 0.17, which is related to the oxygen reduction at the electrode surface [56].

2.3 | Stability Assessment and the Role of Surface Acidity

The PNC-73 coating significantly increases the stability of LSCF air electrodes both in dry and humidified air environments for a long duration of time. Figure 3a shows the stability test of a symmetrical cell consisting of bare LSCF and PNC-73 coated LSCF air electrodes under dry air, 3 v.% H_2O , and 30 v.% H_2O environment at 650°C. The high step change in the R_p value with bare LSCF air is observed from dry air to humidified air environment, specifically at the 50th h, which is very minimal with the case of PNC-73 coated LSCF air electrode. Moreover, the rapid fluctuations in R_p values during stability analysis are also observed during the stability test with the bare LSCF air electrode. The degradation rate of PNC-73 coated LSCF after 200 h under 30 v.% H_2O environment at 650°C is $6.5 \times 10^{-4} \Omega \text{ cm}^2 \text{ h}^{-1}$, which is almost half $12.1 \times 10^{-4} \Omega \text{ cm}^2 \text{ h}^{-1}$ as compared to the bare LSCF. The stability analysis of pure phase PNC air electrode was also conducted to compare with bare LSCF and PNC-73 coated LSCF for 110 h under 3 v.% H_2O environment at 600°C. The results show the rapid fluctuation of R_p values during stability analysis. Figure S15a,b shows the EIS profiles and stability analysis of symmetrical cells with pure phase PNC air electrode, respectively. Figure 3b shows the SEM images of PNC-73 coated LSCF and bare LSCF air electrodes after stability test in 30 v.% H_2O environment. No morphological changes were recorded for the PNC-73 coated LSCF air electrode during the stability test in a 30 v.% H_2O humidified air environment, which demonstrates the robust stability of the reported air electrode in a high-steam environment. However, with bare LSCF, a rough surface was observed, reflecting the effect of high steam on the surface morphology of the bare LSCF air electrode. A similar rough surface was also observed on the pure phase PNC air electrode as given in Figure S16, depicting the effect of high steam on air electrode. The EIS profiles of symmetrical cells with bare LSCF and PNC-73 coated LSCF air electrodes are given in Figure S17 and Figure 3c, respectively. The increase in EIS curves in LF and IF ranges is owing to the deteriorated surface reaction

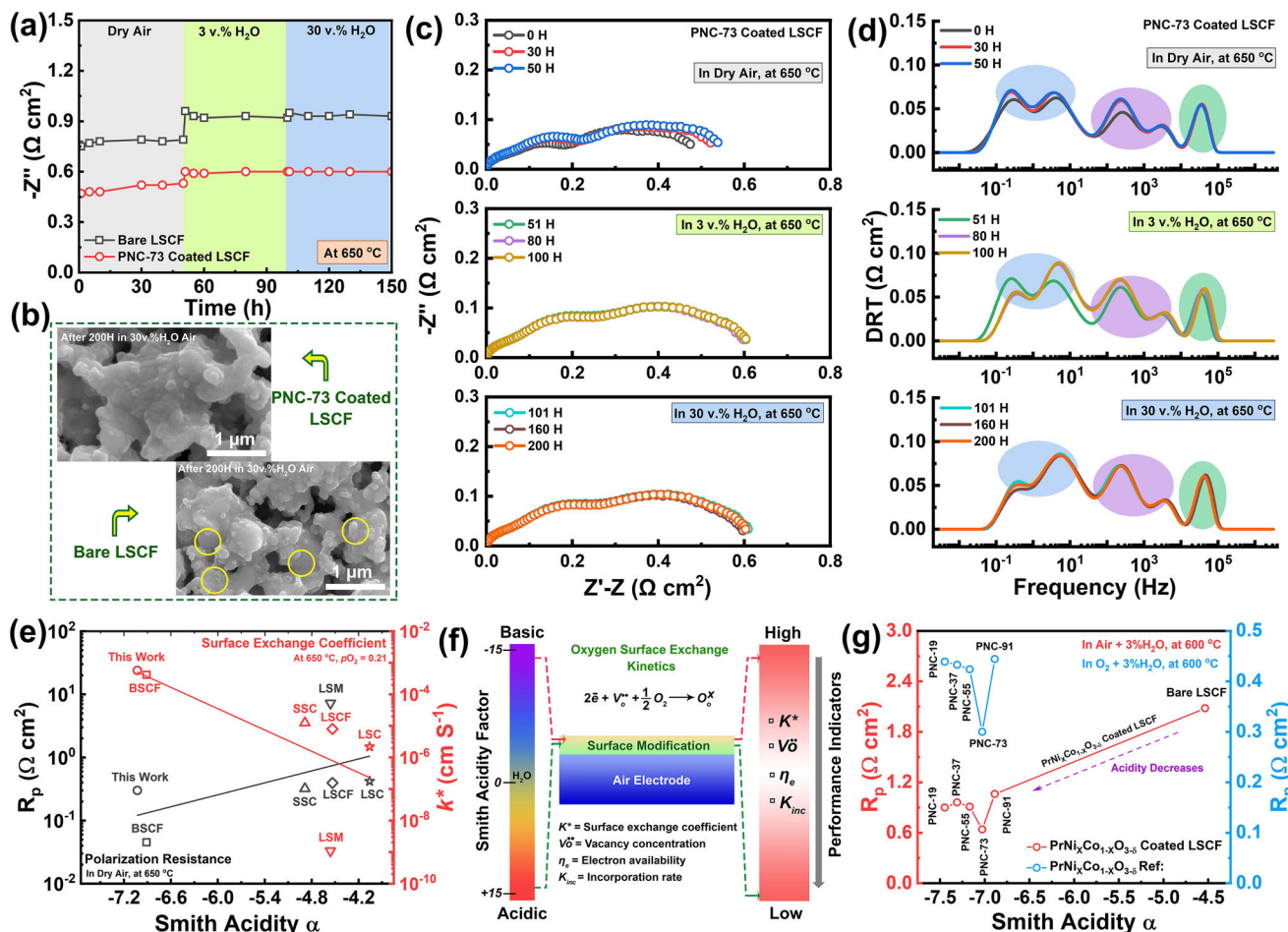


FIGURE 3 | (a) Stability test of symmetrical cells consist of bare LSCF and PNC-73 coated LSCF air electrodes under dry air, 3 v.% H_2O and 30 v.% H_2O environment at 650°C , (b) SEM Images of PNC-73 coated LSCF and bare LSCF air electrode after stability test, (c) Nyquist plots showing the EIS analysis of stability test of symmetrical cell consisting of PNC-73 coated LSCF air electrodes under dry air, 3 v.% H_2O , and 30 v.% H_2O environment at 650°C , (d) DRT profiles of the EIS analysis of PNC-73 coated LSCF air electrodes under dry air, 3 v.% H_2O , and 30 v.% H_2O environment at 650°C , (e) Polarization resistance [71–83] and surface exchange coefficient [33, 84–91] as a function of Smith acidity for previously reported air-electrode materials; and PNC-73 coated LSCF air electrode (measured at 650°C , $p\text{O}_2 = 0.21$ atm), (f) Graphical abstract demonstrating the relationship between the Smith acidity of the surface decorating materials for air electrodes and oxygen surface exchange kinetics, and (g) Polarization resistance as a function of Smith acidity for bare LSCF and $\text{PrNi}_{1-x}\text{Co}_x\text{O}_{3-\delta}$ coated LSCF air electrodes at different compositions, alongside polarization resistance of previously reported PNC-73 air electrodes with varying compositions.

activity, which is more with the bare LSCF and less with PNC-73 coated LSCF air electrodes. The corresponding DRT graph of the EIS profile of PNC-73 coated LSCF air electrode is given in Figure 3d, which also confirms an increase of peaks in LF and IF ranges with some fluctuations with respect to time and change in environment, which is more in case of bare LSCF air electrode. This analysis showed that the PNC-73 coated on LSCF enhanced stability in dry air, humidified air, and high steam environment for a long period of time as compared to the bare LSCF air electrode.

The performance of air electrodes can also be analyzed based on the surface acidity of the air electrode materials, which is derived from the Smith acidity factors of their corresponding binary oxides [65]. The Smith acidity significantly influences the surface exchange kinetics of air electrodes. Numerous studies have reported surface modifications of mixed ionic and electronic conducting (MIEC) oxides using a range of binary oxide materials

with varying acidity factors. These modifications demonstrate the impact of acidity on surface exchange kinetics, leading to enhanced electrochemical performance [66–69]. To quantify the acidity of complex perovskite materials, Nicollet et al. [65] proposed a general expression to determine the Smith acidity factor (α) of perovskite oxides from the acidity factors of their constituent binary oxides. However, Derek S Smith et al. reported the acidity factor values for various binary oxides [70]. The general expression of the Smith acidity factor with formula $\text{A}_x\text{A}'_{1-x}\text{B}_y\text{B}'_{1-y}\text{O}_3$ ($0 \leq x \leq 1$, $0 \leq y \leq 1$) is given in the Equation (1):

$$\alpha_{\text{A}_x\text{A}'_{1-x}\text{B}_y\text{B}'_{1-y}\text{O}_3} = \frac{x\alpha_{\text{AO}} + (1-x)\alpha_{\text{A'O}} + y\alpha_{\text{BO}} + (1-y)\alpha_{\text{B'O}}}{2} \quad (1)$$

Here, x represents the stoichiometric content of A, $(1-x)$ that of A', y the content of B, and $(1-y)$ that of B'. The parameter α denotes the acidity factor of the corresponding oxides A, A', B, and B'.

The surface exchange kinetics are typically analyzed based on the surface exchange coefficient (k^*) and the polarization resistance of the air electrode materials. To investigate the relationship between Smith acidity and these kinetic parameters, average values of R_p and k^* under similar operating conditions for various air electrode materials were extracted from the literature, as presented in Table S2 and Table S3, respectively. Furthermore, the Smith acidity values of the selected air electrode materials from literature and our proposed compositions of $\text{PrNi}_{1-x}\text{Co}_x\text{O}_{3-\delta}$ are calculated using Equation (1) and are provided in Tables S4–S6. Figure 3e shows the relationship between Smith acidity and surface exchange kinetics (k^* and R_p) of air electrode materials reported in literature. Perovskite materials with lower Smith acidity factors exhibit higher surface exchange coefficients and lower polarization resistance, except for the Mn-based perovskite, which does not follow the linear trend observed in the other materials. This relationship between k^* and R_p is significant across a wide range of air electrode materials and has also been reported by many researchers in the literature. Figure 3f illustrates that tuning the acidity of surface-modifying materials influences the surface exchange kinetics of air electrodes. The less acidic modifiers enhance oxygen kinetics and electrode performance, whereas the more acidic modifiers prevent these processes. In this work, we have modified the surface of LSCF with PNC-73 through the solution infiltration method. The Smith acidity of LSCF and PNC-73 are calculated as -4.53 and -7.03 , respectively. As the LSCF is surface modified with low Smith acidity material PNC-73, the surface exchange kinetics (k^* and R_p) of bare LSCF air electrode improved. This reveals that surface modification of the air electrode with materials of lower Smith acidity improves the reaction rates and electrochemical performance. Recently, several researchers, including Nicollet et al. [69], Alexandre et al. [66], and Christoph et al. [67], have also reported the same experimental analysis that modifying the air electrode surface with low Smith acidity materials enhances surface exchange kinetics, which results in an overall improvement in the electrochemical performance of the cells. Figure 3g shows the polarization resistance of $\text{PrNi}_{1-x}\text{Co}_x\text{O}_{3-\delta}$ coated LSCF air electrodes as a function of Smith acidity at 600°C under a 3% H_2O environment. It is observed that modification of LSCF with different compositions of $\text{PrNi}_{1-x}\text{Co}_x\text{O}_{3-\delta}$ leads to a linear decrease in R_p with decreasing Smith acidity. However, the composition PNC-73 deviates from this trend, reducing the R_p of LSCF more significantly compared to the other $\text{PrNi}_{1-x}\text{Co}_x\text{O}_{3-\delta}$ compositions. Although Smith acidity generally correlates with R_p , PNC-73 deviates from this trend in Figure 3g. While PNC-91 exhibits the lowest acidity, PNC-73 gives the best R_p due to the optimal balance of oxygen vacancy concentration (as shown in DFT calculations), PrO_2 phase content (as shown in XRD/TEM analysis), proton transfer barrier (as shown in DFT calculations), and B-site cation synergy ($\text{Ni}/\text{Co} = 7:3$). Ding et al. [29] also reported the similar pattern of R_p reduction with varying $\text{PrNi}_{1-x}\text{Co}_x\text{O}_{3-\delta}$ composition as shown in Figure 3g.

2.4 | Single Cell Performance Analysis

The influence of PNC-73 infiltration on the catalytic activity and durability of the LSCF air electrode is analyzed using Ni-BZCYYb / BZCYYb / LSCF single cells with bare and PNC-coated LSCF air electrodes in both FC and EC modes. In FC mode, the cell

with the PNC-73 coated LSCF delivers the power densities of 0.50 , 0.28 , and 0.15 W cm^{-2} at 700°C , 650°C and 600°C ; respectively, which are higher compared to the cell with bare LSCF air electrode, as shown in Figure 4a,b. At 700°C the cell with the PNC-73 coated LSCF air electrode exhibits 20% increase in power density compared to the bare LSCF air electrode. Moreover, in EC mode at an applied voltage of 1.3 V , the cells with the PNC-73 coated LSCF air electrode achieved current densities of 2.80 , 1.20 , and 0.75 A cm^{-2} at 700°C , 650°C and 600°C ; respectively, as presented in Figure 4c. In comparison, the cell with bare LSCF air electrode shows current densities of 1.33 , 0.91 , and 0.50 A cm^{-2} at the same temperatures as given in Figure 4d. At 700°C in EC mode, the cell with as-reported PNC-73 coated LSCF shows 2.1 times higher current density than the bare LSCF counterpart, indicating the robust catalytic activity of PNC-73 coating in EC mode. The improved electrochemical performances in both FC and EC mode with PNC-73 coating are attributed to its enhanced catalytic activity, resulting from an increase in reaction sites and an extension of the TPB from electrolyte-electrode interface to the bulk air electrode surface. Figure 4e,f shows the comparison of power densities and current densities of a single cell with bare and PNC-73 coated LSCF air electrodes. The EIS profiles of a single cell with bare and PNC-73 coated air electrodes are given in Figure S18 and Figure 4g, respectively. From the EIS analysis, it is also depicted that the improved electrochemical performance of a single cell with PNC-73 coated LSCF is due to the decreased polarization resistance, which is driven by the enhanced catalytic activity. Figure 4h shows the comparison of R_Ω and R_p for single cells with bare LSCF and PNC-73 coated LSCF air electrodes from the temperature range of 700°C to 600°C . The durability of single cells with bare and PNC-73 coated air electrodes is evaluated in both FC and EC modes. As shown in Figure 4i the cell with PNC-73 coated LSCF operates stably for 20 h under cycling conditions between EC mode (1.3 V) and FC mode (0.7 V) and delivering current densities of 0.70 and 0.24 A cm^{-2} at 650°C , respectively. In contrast, the cell with bare LSCF as given in Figure S19, delivers current density of 0.16 and 0.10 A cm^{-2} in EC, and FC modes, respectively, under the same operating conditions. The remarkable 2.1-fold increase in current density in EC mode, combined with robust durability under cycling, underscores the robust catalytic ability of PNC-73 coating and its critical role in optimizing both EC and FC operations.

2.5 | Mechanistic Insights from DFT Calculations

To gain atomic-level insight into the enhanced catalytic activity, we performed DFT calculations to investigate the mechanisms of the OER and ORR on the surface of pristine and defective perovskite PNC structures. Figure 5a shows the energy pathway for the OER (left to right) and the ORR (right to left) on the thermodynamically stable energy PNC-terminated (110) surface. During the electrolysis process, the OER proceeded in the following steps: H_2O adsorption, the dissociation of H_2O to $^*\text{OH}$ and $^*\text{H}$ intermediates. The adsorption of H_2O on the PNC- (110) surface is a spontaneous process with the adsorption energy (-1.2 eV). The $^*\text{OH}$ and $^*\text{H}$ in the presence of water molecules dissociate into the $^*\text{O}^*\text{H}$ with the evolution of one proton. This step is subdivided into two pathways, first the formation of $^*\text{OH}$ by releasing a proton, and then $^*\text{OH}$ transforms

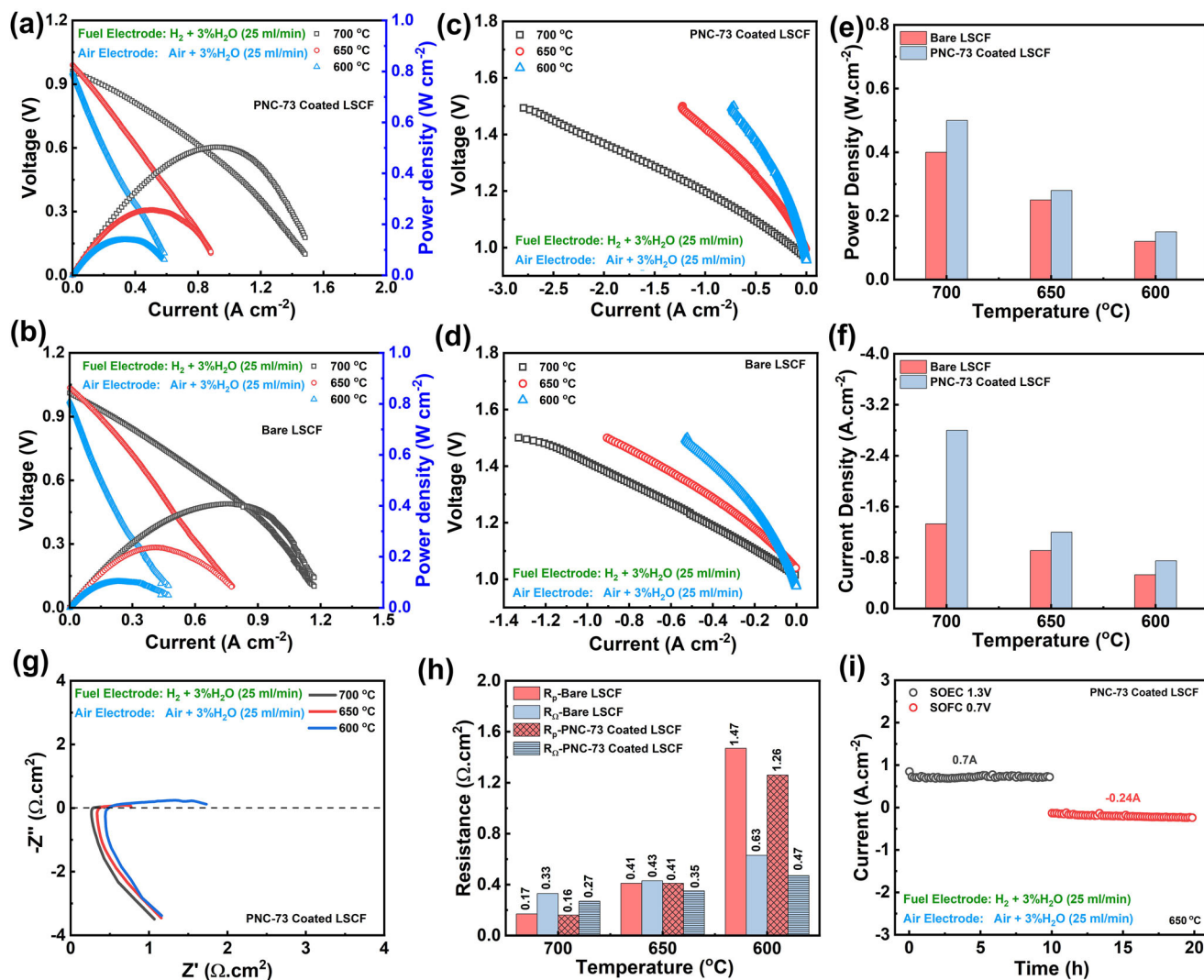


FIGURE 4 | Performance of Ni-BZCYYb/BZCYYb/LSCF single cell with bare and PNC-coated LSCF air electrodes: (a,b) Current-voltage-power (*I-V-P*) curves for cells with bare LSCF and PNC-73 coated LSCF air electrodes, respectively; measured in FC modes at temperature from 700 °C to 600 °C with humidified hydrogen (97% H₂ and 3% H₂O) supplied to the fuel electrode and humidified air (97% Air and 3% H₂O) supplied to the air electrode. (c,d) Current-voltage (*I-V*) curves for cells with bare LSCF and PNC-73 coated LSCF air electrodes, respectively; measured in EC modes at temperature from 700 °C to 600 °C with humidified hydrogen (97% H₂ and 3% H₂O) supplied to the fuel electrode and humidified air (97% Air and 3% H₂O) supplied to the air electrode. (e,f) Comparison of max power densities and current densities for single cells with bare LSCF and PNC-73 coated LSCF air electrodes in FC and EC modes, respectively. (g) EIS profiles of single cell with PNC-73 coated LSCF air electrodes at temperature from 700 °C to 600 °C. (h) Comparison of ohmic resistance (R_o) and polarization resistance (R_p) for single cells with bare LSCF and PNC-73 coated LSCF air electrodes from the temperature range of 700 °C to 600 °C. (i) Stability test of single cells with PNC-73 coated LSCF air electrode in FC and EC modes at 0.7 V and 1.3 V respectively, with 3% H₂O humidified H₂ and 3% H₂O humidified air supplied.

to *O*H, respectively. The formation of *OH required 0.12 eV energy, which is an uphill process with a small energy barrier (0.12 eV), this barrier is lower than the energy barrier (0.58 eV) of *OH dissociation on the PBC surface and that of LSCF (1.19 eV). Second, the formation of *H*O intermediates through an exothermic process with relative energy (−0.43 eV). In the next steps of the OER process, the *O*H intermediates are transformed into *O and to *OOH intermediates respectively, releasing a cascade of protons. The *OOH intermediates release another proton and change to *O, and the combination of the two *O desorbs as an oxygen molecule from the surface. The rate-limiting steps for the OER process are the formation of *OOH and the desorption of the O₂ molecule, with energy barriers 0.32 and 0.35

eV, respectively, which are much lower than the reported values for the rate-determining steps (RDS) of PBC, which are 0.61 and 1.03 eV [32]. In the case of the ORR process, the rate-limiting step is the desorption of H₂O, which has an energy barrier of 1.2 eV, comparable with the ORR barrier on the PBC (1.04 eV) [32]. The small activation energy barriers for OER/ORR on the PNC-(110) surface show that this surface is very feasible for these reactions. The optimized atomic structures of the pristine PNC (110) surface at different stages of the OER/ORR pathways are given in Figure S17. Moreover, we also examined the OER and ORR mechanisms on the A-site defective PNC-(110) surface, with the energy profile shown in Figure 5b (OER: left to right; ORR: right to left). The OER proceeds through: (1) spontaneous H₂O adsorption

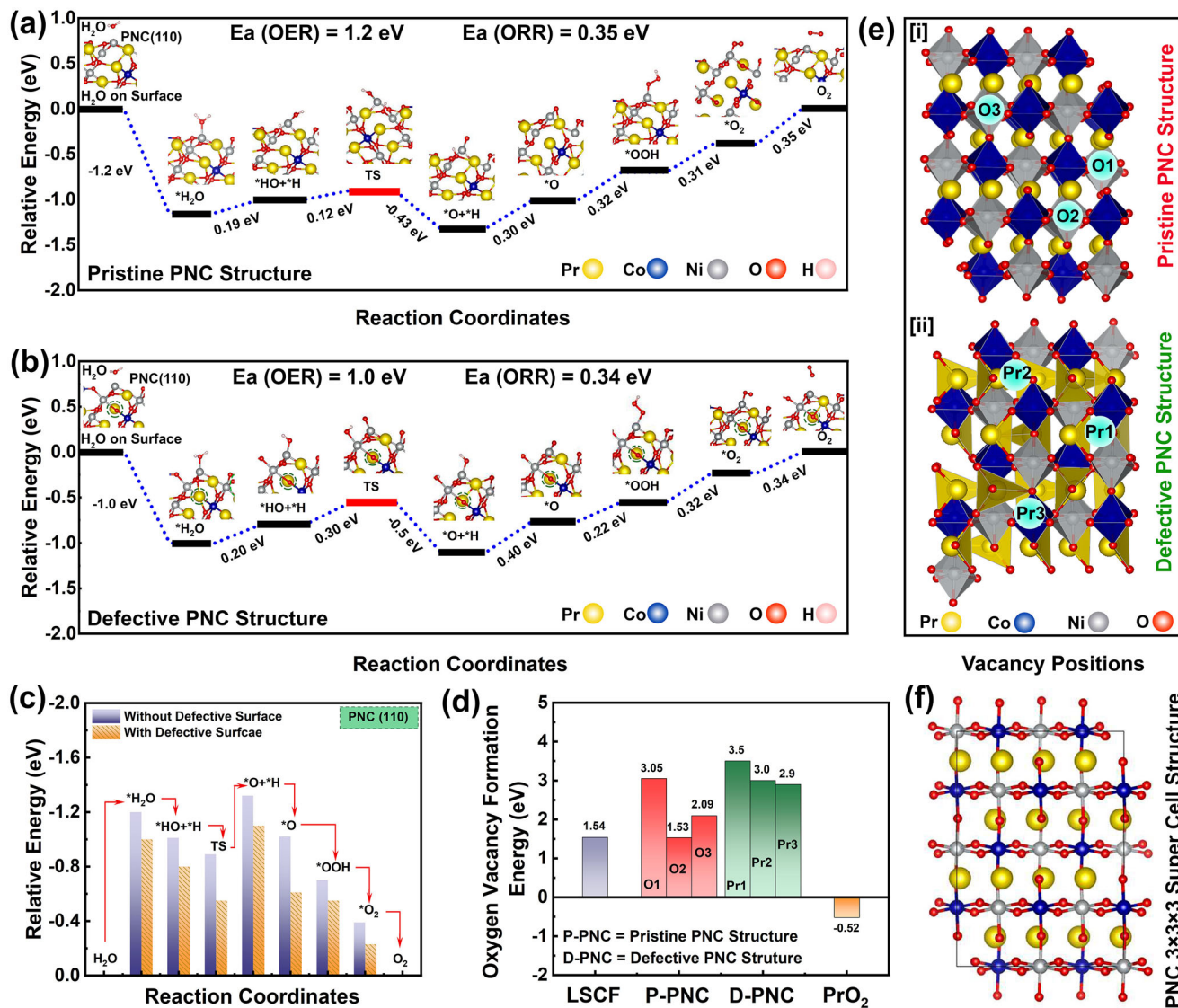


FIGURE 5 | The calculated relative Gibbs free energy for the OER/ORR mechanism on the surface of PNC(110): (a) Pristine PNC surface and (b) Defective PNC surface, (c) Comparison of the Gibbs free energy for the OER/ORR mechanism on the pristine and defective surface of PNC, (d) Comparison of calculated vacancy formation energies in pristine PNC structure at the three distinct oxygen sites (O1, O2, O3) and defective PNC structure for the three distinct Pr sites (Pr1, Pr2, Pr3), along with vacancy formation energies of LSCF and PrO₂ as reported in recent literature, (e) Polyhedral views of the 3 × 3 × 3 super cell (space group *Pnma*) used for the calculations for the pristine PNC structure and the defective PNC structure containing a vacancy, and (f) the atomic structure of PNC 3 × 3 × 3 super cell structure used for DFT calculations.

(−1.2 eV), (2) dissociation of *H₂O into *OH and *H (where * denotes adsorbed species) with a 0.30 eV barrier, (3) exothermic formation of *OH (−0.50 eV), (4) transformation of *OH to *O and then *OOH with proton release, and (5) combination of two *O to desorb O₂. The rate-limiting steps are *O formation (0.40 eV) and O₂ desorption (0.34 eV), both lower than PBC's barriers (0.61 and 1.03 eV) [32]. For ORR, the rate-limiting step is H₂O adsorption (1.0 eV), comparable to PBC (1.04 eV). These low barriers highlight the A-site defective PNC(110) surface's suitability for efficient OER/ORR catalysis. The optimized atomic structures of the defective PNC (110) surface at different stages of the OER/ORR pathways are given in Figure S18. The comparison of the Gibbs free energy for the OER/ORR mechanism on the pristine and defective surface of PNC is shown in Figure 5c, which depicts that the defective PNC exhibits lower energy pathways

and results in reduced overpotential and enhanced catalytic activity.

The oxygen vacancy formation energy is calculated according to the following formula; $E_f = E_{\text{defect}} + \frac{1}{2}E_{\text{O}_2} - E_{\text{perfect}}$. Where E_f is the oxygen vacancy formation energy, E_{defect} , E_{O_2} and E_{perfect} are the energy of the supercell with an oxygen vacancy, energy of a free oxygen molecule, and energy of the pristine PNC supercell, respectively. Similarly, the A-site formation energy is calculated according to the formula; $E_f = E_{\text{defect}} + \frac{1}{2}E_{\text{Pr}} - E_{\text{perfect}}$. Where E_f , E_{defect} , E_{Pr} and E_{perfect} is the A-site vacancy formation energy, energy of the PNC with defect, energy of the Pr atom, and energy of the pristine PNC; respectively. We considered three different sites (O1, O2, and O3) for the oxygen vacancy formation energy calculations. The vacancy formation energy of oxygen for the

three sites is 3.05, 1.53, and 2.096 eV, respectively. The oxygen at the O2 lattice site has a lower formation energy (1.53 eV) than the O1 and O3. This reduced formation energy indicates that the oxygen vacancies are most likely to form at this site. The lower energy may be attributed to differences in local bonding, such as weaker Ni-O or Pr-O interactions at the O2 site, which facilitate the removal of an oxygen atom. In contrast, the higher formation energy at the O1 site (3.05 eV) indicates greater stability of the oxygen atom in that position, likely due to stronger coordination with neighbouring cations. Similarly, the calculated A-site vacancy formation energies are 3.5, 3, and 2.9 eV for Pr1, Pr2, and Pr3 sites, respectively. These results suggest moderate variability in the formation energy across different A-site positions or configurations. The lowest formation energy (2.9 eV) indicates that A-site vacancies are relatively more favorable at certain sites, potentially due to differences in local strain or electrostatic interactions within the perovskite lattice. The higher formation energy of 3.5 eV at another site suggests stronger bonding or greater lattice stability, making vacancy formation less favorable. Figure 5d shows the comparison of calculated oxygen vacancy formation energies in the pristine PNC structure at the three distinct oxygen sites (O1, O2, O3) and defective PNC structure for the three distinct Pr sites (Pr1, Pr2, Pr3); along with the oxygen vacancy formation energies of LSCF and PrO_2 as reported by Yao et al. [42] in recent work. The -0.52 eV value for PrO_2 , reported by Yao et al. [42], indicates exothermic vacancy formation relative to the reference states used in their DFT framework, though the absolute sign is reference-state dependent. The key comparative trend is that PrO_2 exhibits a substantially lower E_f than LSCF and PNC perovskite, confirming that oxygen vacancies form most readily in the PrO_2 phase. Figure 5e shows the polyhedral views of the $3 \times 3 \times 3$ supercell (space group $Pnma$) used for the calculations for the pristine PNC structure and the defective PNC structure containing a vacancy. The oxygen vacancy formation energies of both pristine and defective PNC at different lattice sites are higher compared to those of LSCF, except for the O2 site in pristine PNC, which shows a slightly lower value than LSCF but with an insignificant difference. However, the vacancy formation energy of PrO_2 is much lower than the LSCF and PNC (pristine and defective at various sites). As observed from the HR-TEM image, phase structures of PNC-73 coated LSCF air electrode, and XRD profiles of the co-fired PNC-73 material, the PrO_2 phase emerges. The appearance of PrO_2 phase after co-firing the PNC-73 coated LSCF is the reason for accelerating the catalytic reactions. Furthermore, the EIS profiles of symmetrical cells based on PNC-73 infiltrated LSCF air electrodes depicted the much lower R_p values and lower activation energy compared to LSCF air electrodes, indicating improved ORR kinetics and enhanced electrochemical performance. Figure 5f the atomic structure of the PNC $3 \times 3 \times 3$ super cell structure used for DFT calculations.

3 | Conclusion

In summary, we have demonstrated a highly effective and scalable surface-engineering strategy to overcome the major limitations of conventional LSCF air electrodes in reversible protonic ceramic cells (R-PCCs). Through the optimized infiltration of a $\text{PrNi}_{0.7}\text{Co}_{0.3}\text{O}_{3-\delta}$ (PNC-73) catalyst, we successfully transformed a mixed ionic-electronic conductor into a high-performance

triple-conducting electrode. The key achievement of this work is the creation of a dual-phase nano-coating, comprising the PNC-73 perovskite and a PrO_2 phase, which acts synergistically to enhance both catalytic activity and durability. The PNC-73-coated LSCF electrode delivered exceptional electrochemical performance, achieving a remarkably low polarization resistance of $0.30 \Omega\text{cm}^2$ at 650°C – a four-fold improvement over bare LSCF. Crucially, the modified electrode exhibited outstanding operational stability, with a degradation rate nearly half that of its unmodified counterpart under a high-steam (30% H_2O) environment, addressing a critical barrier to commercialization. The performance enhancement is attributed to two primary mechanisms: (1) the extension of proton conduction pathways from the triple-phase boundary to the entire electrode surface, vastly increasing the number of active sites, and (2) the introduction of abundant oxygen vacancies via the PrO_2 phase, which dramatically accelerates surface oxygen exchange kinetics. These mechanisms were corroborated by DRT analysis, ECR measurements, and supported by DFT calculations, which revealed low energy barriers for oxygen electrocatalysis on the PNC surface. This work provides more than just a superior electrode material; it establishes a generalizable paradigm for revitalizing well-established but limited electrode materials by imparting multifunctional triple-conducting capability via a simple and cost-effective infiltration process. The insights linking the PNC-73 coating's low Smith acidity to enhanced surface kinetics offer a valuable design principle for future catalyst development. By successfully decoupling ionic transport functionality from the bulk electrode scaffold, this approach paves the way for the development of highly active, durable, and commercially viable electrodes for next-generation intermediate-temperature solid-state electrochemical devices.

4 | Experimental Section

4.1 | Materials Synthesis

4.1.1 | Catalyst Precursor Solution Preparation

The catalyst solution for the infiltration of the LSCF air electrode was prepared by dissolving the metal nitrates in distilled water at a concentration of 0.1 mol L^{-1} . The materials used as the metal source for the infiltrate solution were $\text{Pr}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$ (99.9%, Alfa-Aesar), $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (99%, Sigma-Aldrich), and $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (99%, Sigma-Aldrich). The stoichiometric amounts of metal nitrates for the preparation of $\text{PrNi}_x\text{Co}_{1-x}\text{O}_{3-\delta}$ (where $x = 0.1, 0.3, 0.5, 0.7$, and 0.9) catalyst compositions were dissolved in a solvent consisting of distilled water and ethanol in a volume ratio of 4:1. Polyvinyl pyrrolidone (PVP) and glycine were added in equal ratio as a surfactant and a complexing agent, respectively.

4.1.2 | Fabrication of LSCF Air Electrode

The LSCF ($\text{La}_{0.6}\text{Sr}_{0.4}\text{Co}_{0.2}\text{Fe}_{0.8}\text{O}_{3-\delta}$) air electrodes were prepared using the tape-casting method. The air electrode slurry for tape casting was prepared in two steps. In the first step, LSCF powder, a pore-forming agent (graphite, 6%), solvents (ethanol and n-butyl acetate, 30%), and a dispersant (fish oil, 6%) were ball milled

for 24 h. In the second step, plasticizers (butyl benzyl phthalate, 3% and polyethylene glycol, 5%) were added and ball milled for another 24 h, followed by the addition of a binder (polyvinyl butyral, 6%) and an additional 6 h of ball milling. The prepared slurry for tape casting was then coated on PET (polyethylene terephthalate) sheets using a slide coating machine and dried overnight at room temperature. The dried LSCF-coated PET sheet was then punched into discs of 0.28 cm diameter, and the average mass for each LSCF air electrode disc was 4 mg.

4.1.3 | Synthesis of BZCYYb Electrolyte Powder

The $\text{BaZr}_{0.1}\text{Ce}_{0.7}\text{Y}_{0.1}\text{Yb}_{0.1}\text{O}_{3-\delta}$ (BZCYYb) electrolyte powder was synthesized using the solid-state reaction method. Stoichiometric amounts of barium carbonate (BaCO_3), zirconium oxide (ZrO_2), cerium oxide (CeO_2), yttrium oxide (Y_2O_3), and ytterbium oxide (Yb_2O_3) powders were ball-milled in ethanol at 400 rpm for 24 h. The resulting electrolyte powder was then pelletized and sintered at 1100°C for 10 h in an air environment. The ball milling and sintering processes were repeated three times. Next, 1% NiO was added as a sintering aid to the electrolyte powder, followed by ball milling for another 24 h and sintering at 1100°C for 10 h. Polyvinyl butyral (PVB) binder was added to the finalized BZCYYb powder, which was then ball-milled again for 6 h and ground to a fine powder to prepare electrolyte pellets for a symmetrical cell.

4.2 | Cell Fabrication

4.2.1 | Symmetrical Cells

The BZCYYb electrolyte pellets for symmetrical cells were prepared using the dry pressing method and sintered at 1450°C for 5 h to obtain dense pellets with a diameter of 3 cm. For the preparation of LSCF|BZCYYb|LSCF configured symmetrical cells, a solution of SDC buffer layer was dropped onto the dense electrolyte pellets, followed by the placement of the as-prepared tape-cast LSCF air electrode on both sides of the BZCYYb electrolyte and drying in an oven at 80°C for 5 h. The LSCF|BZCYYb|LSCF configured symmetrical cells were co-fired at 1100°C for 2 h to attach the air electrode to the electrolyte and to create a porous skeleton of the LSCF air electrode with an active area of 0.28 cm^2 . For the modification of the LSCF air electrode, appropriate amounts (5, 10, or $15\text{ }\mu\text{L}$) and compositions ($\text{PrNi}_x\text{Co}_{1-x}\text{O}_{3-\delta}$) of the as-prepared catalyst solution were infiltrated into the LSCF skeleton and co-fired at 700°C for 2 h in an air environment. The catalyst load for $1\text{ }\mu\text{L}$ was $12.4\text{ }\mu\text{g}$, assuming 100% precursor conversion to oxide, while the ratio of catalyst mass to LSCF skeleton mass was calculated as 1.53 wt.%.

4.2.2 | Anode-Supported Single Cells

The fuel-electrode supported single cells consisted of five layers: (1) fuel-electrode supported layer-1 (55% NiO + 45% BZCYYb) with thickness of 40–45 μm , (2) fuel-electrode supported layer-2 (50% NiO + 50% BZCYYb) with thickness of 280–310 μm , (3) fuel-electrode functional layer (55% NiO + 45% BZCYYb) with thickness of 40–45 μm , (4) electrolyte layer (BZCYYb) with thickness of 10–15 μm , and (5) air electrode (65% LSCF + 35%

BZCYYb). The air electrode slurry composed of LSCF+BZCYYb and α -terpineol (6 wt% ethyl cellulose) was mixed manually in equal ratios for 45 min and uniformly brush-painted onto the fuel-electrode supported pellets; and drying at 80°C for 5 h, followed by sintering at 1000°C for 2 h. For the modification of the LSCF air electrode, $5\text{ }\mu\text{L}$ of PNC-73 catalyst solution was infiltrated onto the LSCF air electrode (active area = 0.28 cm^2) and co-fired at 700°C for 2 h in a box furnace.

4.3 | Characterization and Electrochemical Measurements

4.3.1 | Structural and Morphological Characterization

The surface morphology and elemental mapping of air electrodes were carried out using scanning electron microscopy-SEM (Hitachi S3400N, FESEM), transmission electron microscopy-TEM (FEI Tecnai F30 super-twin TEM with super twin objective lens), and high-resolution transmission electron microscopy-HRTEM (Gatan One-View camera). The X-ray diffraction analysis was performed using a Rigaku MiniFlex 600 with Cu $K\alpha$ radiation ($\lambda = 1.5406\text{ \AA}$) at a scanning rate of $10^\circ\text{C min}^{-1}$, with an operating voltage of 40 kV and a current of 15 mA. Moreover, the GSAS-II software was used for Rietveld refinement of the XRD data. The XPS measurements were performed on a Thermo Scientific K-Alpha+ spectrometer under ultrahigh vacuum ($< 2.0 \times 10^{-7}$ mbar) using a 400 μm monochromated Al $K\alpha$ source (12 kV, 6 mA). Survey spectra were recorded with a pass energy of 150 eV and a step size of 1.0 eV, while high-resolution core-level spectra were acquired with a pass energy of 50 eV and a step size of 0.10 eV. All spectra were referenced to adventitious carbon (C 1s = 284.8 eV) to correct for charging effects. The EPR measurements were performed on a Bruker EMXnano spectrometer operating at X-band (range from 9.2 to 9.7 GHz). The measurements were performed at room temperature using a microwave power of 20 mW and a modulation frequency of 100 kHz with an amplitude of 1 G.

4.3.2 | Electrochemical Characterization

For electrochemical performance measurements, two-electrode symmetrical cells (LSCF|BZCYYb|LSCF) were fabricated. The area-specific resistance (ASR) was measured using a silver wire mesh as the current collector on both sides of the cells, within a temperature range from 450°C to 650°C in 50°C increments, under a dry air and 3 v.% H_2O environment. The R_p was calculated from the electrochemical impedance spectra (EIS) by determining the difference between the low and high-frequency intercepts on the real axis. The EIS was measured using an Autolab electrochemical workstation (Metrohm Autolab-M204 Model) with an AC amplitude of 50 mV in the frequency range of 1 MHz to 0.01 Hz.

For the electrochemical performance analysis of single cells, button cells configured as Ni-BZCYYb|BZCYYb|LSCF (both bare LSCF and PNC-73 coated LSCF) were employed to measure the I-V curves. These measurements were taken at operating temperatures ranging from 700°C to 600°C in both fuel cell and electrolysis modes. Hydrogen gas containing 3 v.% H_2O vapor

was supplied to the fuel electrode, while the air electrode was exposed to 3 v.% H₂O humidified air. The flow of H₂ and air to the fuel electrode and air electrode was fixed to 25 ml/min for all experimental analysis.

4.3.3 | Distribution of Relaxation Times Analysis

The distribution of relaxation time computation was performed using Bayesian ridge regression approach (also known as Tikhonov regularization approach) without adjusting the parameters as described in previous studies. This method effectively removes pseudo peaks and captures discontinuities in the DRT, thereby enabling the identification of the types and quantities of processes in an electrochemical cell. For ease of use, a Matlab based graphical user interface named as “DRTtools” (a freely available Matlab toolbox for DRT computation) was used in this work as reported in previous studies [92, 93].

4.3.4 | Electrical Conductivity Relaxation

For ECR measurements, dense bars of LSCF material were prepared through the dry pressing method at 20 MPa pressure, followed by sintering at 1250°C for 6 h. The surfaces of the sintered LSCF bars were polished manually using silicon carbide attrition paper (mesh 1200) and cleaned in ethanol using ultrasonic treatment for 30 min. The final dimensions of the polished LSCF bars were recorded as 17.5 × 5.5 × 5.5 (L × W × H) mm³. For the preparation of catalyst-modified LSCF bars, a thin layer (15 nm) of PNC-73 catalyst solution was drop-coated onto the as-prepared LSCF bars and co-fired at 700°C for 2 h. The thickness of the catalyst coating layer was calculated using the Equation (2):

$$L = \frac{VCM}{\rho S} \quad (2)$$

where **L** is the thickness of the coating layer, **V** is the volume of catalyst solution, **C** is the solution concentration, **M** is the molar weight of catalyst solution, **ρ** is the density of catalyst solution, and **S** is the surface area of the LSCF bar. The silver wires were used as the conductors for *I* source and *V* sense and connected with LSCF bars with a 3D (2D model) connection. The silver paste was also used with silver wires to ensure 100% contact of silver wires with LSCF bars for accuracy.

The ECR measurements were conducted to evaluate the surface exchange kinetics of the bare LSCF and PNC-73 coated LSCF. The electrical conductivity was calculated using the 4-probe DC method, in which the constant current of 10 mA was applied to the two silver wires connect with LSCF bar, and the voltage was recorded from the other two silver wires using KEITHLEY source meter (Model-2401). The conductivity measurements were observed over a temperature range from 550°C to 650°C at intervals of 50°C. However, the samples were heated to each respective operating temperature for 1 h before the conductivity testing to bring the system to an equilibrium state. In ECR measurements, a step change in the partial pressure of oxygen (*p*O₂) at a constant temperature is employed to measure the total conductivity of the sample. This method was also applied to both bare and PNC-73 coated LSCF samples. The initial partial

pressure of the oxygen was set to 0.2 atm, and the step change in *p*O₂ was set to 0.1 atm; which is half with respect to initial *p*O₂. However, the overall air flow rate was 100 ml/min. The relaxation time for the conductivity of the samples to reach the equilibrium state was set to 5000 s. All calculations were performed using a freely available Matlab toolbox called ECRtools [94]. This toolbox also facilitates the computation of the normalized conductivity as well as the *K*_{chem}, as documented in prior studies.

4.4 | Computational Methods

We performed spin-polarized DFT calculations with the plane wave code, Vienna Ab-initio Simulation Package (VASP) [95, 96]. The electrons and ions interaction was taken into account by using the projector augmented wave (PAW) method, and the exchange-correlation functional was treated within the generalized gradient approximation (GGA) parameterized by Perdew–Burke–Ernzerhof (PBE) [95]. For the energy calculations, we employed a constant plane-wave kinetic energy cutoff of 400 eV. To account for the strong electron correlations in the 3d valence states of Cobalt (Co) and Nickel (Ni), we employed the DFT+U approach, implementing the Hubbard *U* correction within the simplified, rotationally invariant formalism of Dudarev et al. on the PBE functional [97]. In the GGA+U method, the key parameter is the value of the effective potential, *U*_{eff} = *U*–*J*. The values were set to *U*_{eff} = 3.3 eV for Co and 6.4 eV for Ni, following a previous study [3]. All calculations, including both standard GGA-PBE and GGA+U, were spin-polarized. For Brillouin zone integration, we employed Monkhorst-Pack *k*-point meshes with 3 × 3 × 3 for bulk structure optimization, while a denser mesh of 11 × 11 × 1 was used for surface calculations [98]. The (110) surface was modeled by cleaving a slab with a thickness of approximately 2.5 Å from the optimized bulk structure of Pr₄Ni₂Co₂O₁₂ (space group: *Pnma*) known as PNC. We employed a 3 × 3 × 1 supercell to the (110) truncated surface of PNC, which contains a total of 58 atoms. A vacuum layer of 15 Å was added along the *c*-direction to prevent spurious interactions between periodic images of the slab. During geometry optimization, a selective dynamics approach was applied; the atoms in the top layer were allowed to relax, while the atoms in the bottom layer were fixed to their bulk positions to mimic the subsurface constraints.

Author Contributions

Yinghua Niu and Muhammad Waqas performed experiments, data acquisition, analysis of results, acquired the funding, and draft the article. Mengjun Tang performed formal analysis and investigation. Shafiq Ur Rehman conducted the DFT calculations and theoretical analysis. Sheng Ma assisted in arranging the experimental setup. Modeste Venin Mendieev Nitou and Bismark Boateng revised the research article. Weiqiang Lv supervised the work, acquired the funding, provided lab resources, and revised the final draft.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data supporting this article have been included as part of the Supplementary Information (SI).

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File: sml175308-sup-0001-SuppMat.docx.